Google Cloud H2S

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.

Participant Details

Participant Name: Piyush Kumar Jayant

Problem Statement: Build a production-ready multi-agent

Brief about the idea

Multi-Agent Productivity Assistant is a cloud-native GenAI productivity workspace that unifies tasks, notes, calendar events, and analytics in one conversational experience.

A coordinator agent routes user requests to specialized task, notes, calendar, and analytics agents. AlloyDB AI powers semantic note retrieval, BigQuery powers productivity reporting, and Cloud Run makes the solution publicly deployable for real users and live demos.

Your solution should be able to explain the following:

- How did you approach the common problem statement and translate it into a working solution using ADK, MCP/AlloyDB AI while ensuring Hackathon's core requirements are met?
- What real-world problem does your solution address, and what practical impact does it create for users?
- What is the core approach or workflow behind your solution?

We translated the hackathon stack into a working solution by using Google ADK for orchestration, MCP for safe tool access, AlloyDB for operational data, AlloyDB AI for semantic note search, and BigQuery MCP for analytics.

The product solves fragmented personal productivity workflows. Instead of switching between separate task, notes, calendar, and reporting tools, users can manage work from one assistant.

Core workflow: user request -> root coordinator -> specialized agent -> MCP tool execution -> AlloyDB or BigQuery response -> concise user answer.
Prototype mode also supports analytics-only demos for faster deployment.

Opportunities

- How different is it from any of the other existing ideas?
- USP of the proposed solution

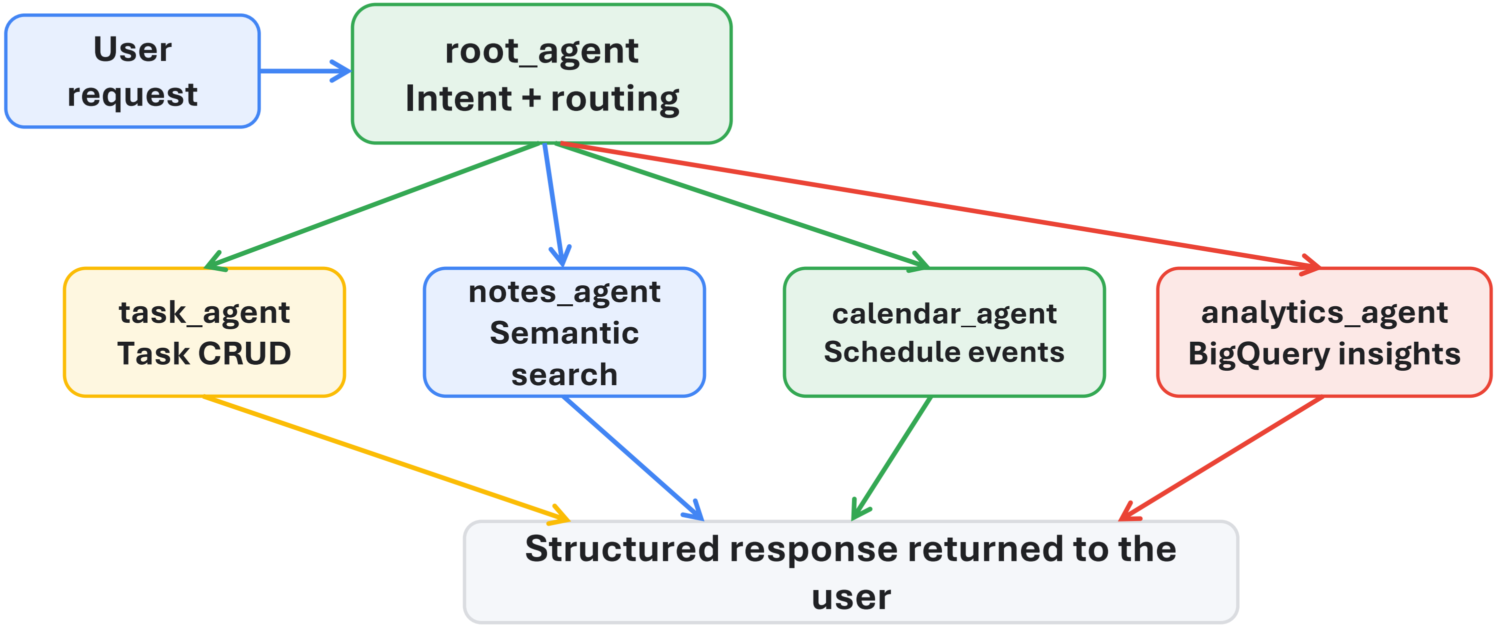
- One assistant handles four productivity jobs instead of isolated point solutions.
- MCP separates reasoning from database access, which improves maintainability and trust.
- AlloyDB AI generates embeddings in-database, reducing extra infrastructure and latency.
- BigQuery analytics turns operational data into measurable productivity insights.
- Cloud Run deployment and prototype mode make the solution demo-ready and scalable.

List of features offered by the solution

- Natural-language task management: create, list, update, and delete tasks
- Semantic note search with AlloyDB AI embeddings using text-embedding-005
- Calendar scheduling for events, meetings, and reminders
- Productivity analytics for completion rates, trends, and activity summaries
- Multi-step request handling across multiple specialized agents
- Public Cloud Run deployment with Vertex AI-backed Gemini routing
- Declarative MCP toolbox layer for clean database operations
- Full mode and prototype mode for rich workflows or fast demos

Process flow diagram or Use-case diagram

End-to-end workflow from user request to action execution and response.



Wireframes/Mock diagrams of the proposed solution

Concept wireframe showing the unified assistant experience.

Conversation Workspace

User:
Create a high-priority review task for Friday.

Assistant:
Task created.

Quick actions
- Add note
- Schedule meeting
- Show analytics

Tasks + Notes

1. Demo rehearsal
2. Investor FAQ
3. Notes about presentation

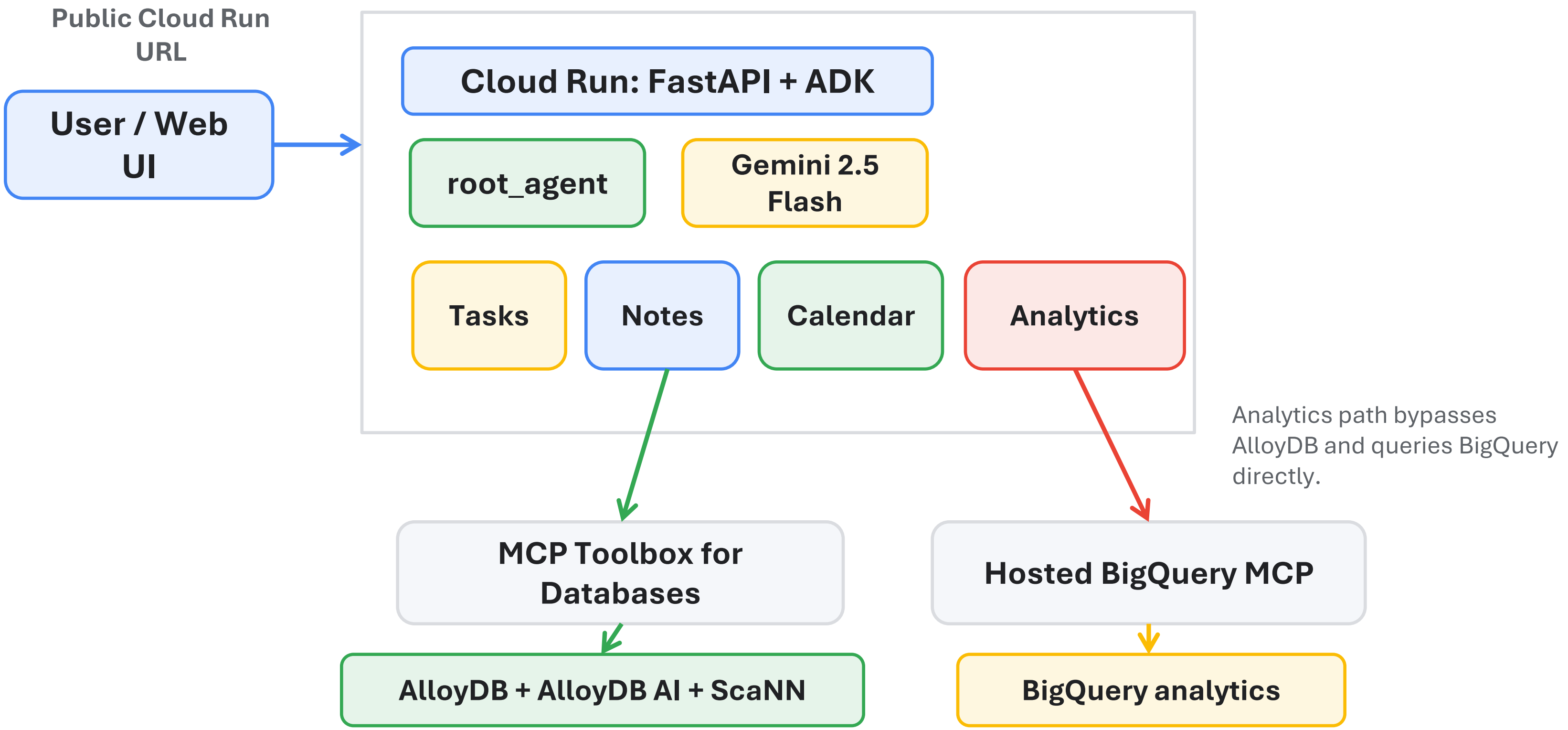
Calendar

09 Apr 2:00 PM
Team sync
45 min

Analytics Panel

Weekly completion rate: 78%
Best day: Tuesday
Most active category: high-priority tasks

Architecture diagram of the proposed solution:



Technologies/Google Services used in the solution

(Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

Google ADK
- Multi-agent orchestration with a root coordinator and specialized sub-agents

Gemini 2.5 Flash via Vertex AI
- Fast reasoning, enterprise-friendly access pattern, and Cloud Run compatibility

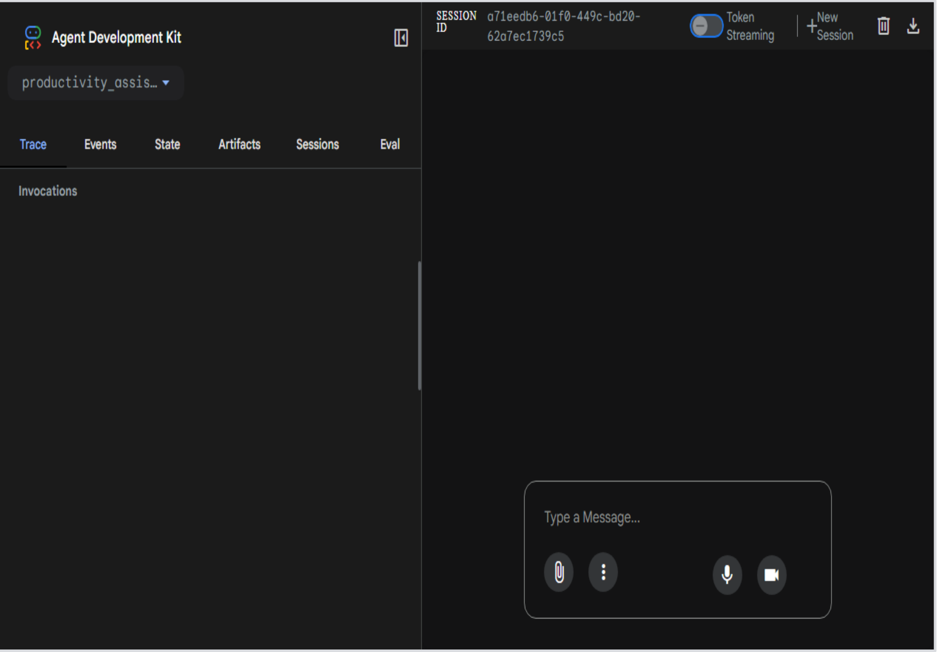
Cloud Run
- Public deployment endpoint, serverless scaling, and simple demo operations

MCP Toolbox for Databases
- Safe, declarative SQL tools for tasks, notes, and calendar operations

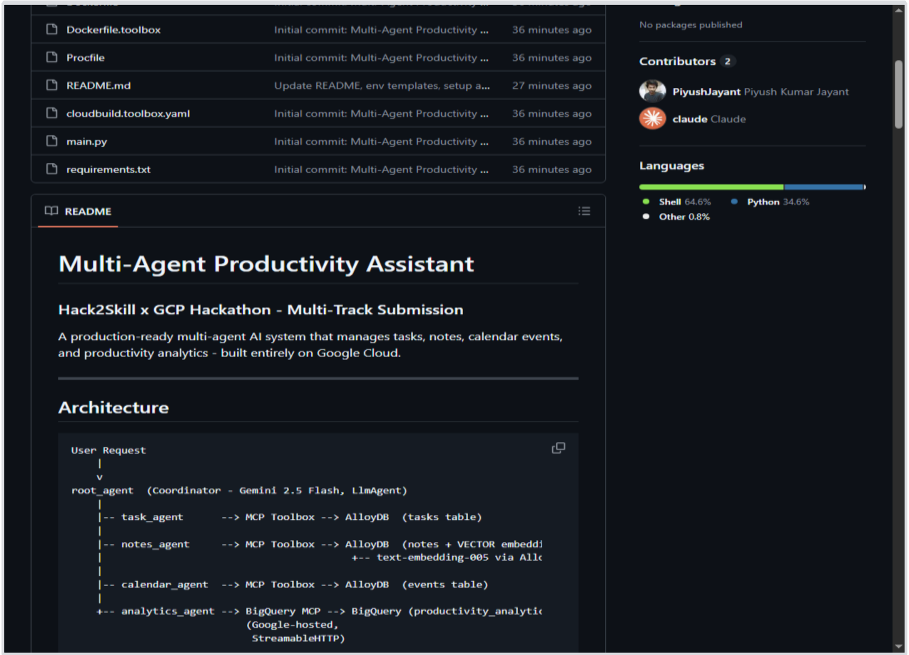
AlloyDB + AlloyDB AI + pgvector / ScaNN
- Operational store plus in-database semantic embeddings and fast similarity search

BigQuery MCP + BigQuery
- Analytics queries over productivity datasets with minimal custom backend code

Snapshots of the prototype



Live Cloud Run deployment
Public ADK interface with agent list available



Public GitHub repository
README documents architecture and services used

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Thank you